

# The p-Bit as a Geometric Tipping Point

From the  $\xi$  Length Hierarchy to the Torsion Barrier  
and the Resolution of the Stochastic Interpretation

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<https://github.com/jpascher/T0-Time-Mass-Duality>

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## .1 Introduction: The Operating Point Between Bit and Qubit

Doc. 178 established the stability of spin as macroscopic evidence for discrete torsion configurations. The permanent magnet shows: spin-up and spin-down are geometrically distinguished states, robust over years without cooling. The classical bit uses precisely this regime — a deep potential well  $U \gg k_B T$ .

The qubit operates at the opposite extreme: metastable torsion, coherence in the millisecond range (Doc. 174), enforced by extreme cooling to mK temperatures.

The **p-bit** (probabilistic bit) occupies the conceptually hitherto unexplored intermediate range: a magnetic nanostructure of the order of the geometric scale  $L_8 \approx 22$  nm of the  $\xi$ -hierarchy, at which the torsion barrier becomes thermally surmountable.

### The central thesis of this document

The thermal noise of the p-bit is not fundamental randomness. It is the macroscopically unresolved sampling of deterministic torsion configurations. The apparent stochasticity is epistemic — the transitions follow the sub-Planck-scale geometry operating on the timescale  $t_0 = t_P/7500$ .

The physical barrier is given materially by the Stoner-Wohlfarth energy  $U = K_u \cdot V$ . The FFGFT identifies: (a) **where** natural p-bit candidates are geometrically to be found (scale  $L_N$ ), and (b) **how** the apparent randomness is to be understood physically.

## .2 The $\xi$ -Length Hierarchy (Review of Doc. 173)

Doc. 173 defines the geometric scaling ladder of the FFGFT via the parameter  $\xi_{\text{par}} = 4/30000 \approx 1,333 \times 10^{-4}$ :

$$L_N = L_0 \cdot \left( \frac{1}{\xi_{\text{par}}} \right)^N, \quad L_0 = \xi_{\text{par}} \cdot \ell_P, \quad (1)$$

where  $\ell_P = 1,616 \times 10^{-35}$  m is the Planck length. Level  $N = 8$  gives:

$$L_8 = \ell_P \cdot \xi_{\text{par}}^{-7} \approx 21,6 \text{ nm}. \quad (2)$$

This scale is identified in Doc. 173 as the universal geometric anchor: excitonic Bohr radius, fine structure constant, transistor threshold size — all at  $N \approx 8,00$ .

## .3 From Length Scale to Characteristic Energy

### Torsion field characteristic at scale $L_N$

Each geometric scale  $L_N$  is assigned a characteristic torsion energy in the FFGFT framework. The torsion field is a relativistic field object; the natural energy scale of a torsion mode at wavelength  $L_N$  is:

$$E_N \equiv \frac{\hbar c}{L_N} = \frac{\hbar c}{L_0} \xi_{\text{par}}^N = \frac{E_P}{\xi_{\text{par}}} \xi_{\text{par}}^N = E_P \cdot \xi_{\text{par}}^{N-1}, \quad (3)$$

where  $E_P = \hbar c/\ell_P = \hbar/t_P \approx 1,956 \times 10^9$  J is the Planck energy. Since  $\xi_{\text{par}} < 1$ ,  $E_N$  decreases strictly monotonically with increasing  $N$ ; each level is a factor of  $1/\xi_{\text{par}} = 7500$  smaller than the previous one.

**Verification:  $N = 1$  yields the Planck energy**

For  $N = 1$ :

$$E_1 = E_P \cdot \xi_{\text{par}}^0 = E_P \approx 1,956 \times 10^9 \text{ J.} \quad (4)$$

The Planck energy is the torsion energy of the Planck level — not a free parameter, but a geometric consequence of the hierarchy.

**Complete energy hierarchy and the p-bit regime**

| $N$      | $L_N$                                             | $E_N = \hbar c / L_N$                              | $E_N / k_B T_{300}$  | Physical domain                                   |
|----------|---------------------------------------------------|----------------------------------------------------|----------------------|---------------------------------------------------|
| 0        | $2,16 \times 10^{-39} \text{ m}$                  | $1,47 \times 10^{13} \text{ J}$                    | $3,5 \times 10^{33}$ | sub-Planck ( $L_0$ )                              |
| 1        | $1,62 \times 10^{-35} \text{ m}$                  | $1,96 \times 10^9 \text{ J}$                       | $4,7 \times 10^{29}$ | Planck scale; $E_1 = E_P$                         |
| 2        | $1,21 \times 10^{-31} \text{ m}$                  | $2,61 \times 10^5 \text{ J}$                       | $6,3 \times 10^{25}$ | —                                                 |
| 3        | $9,09 \times 10^{-28} \text{ m}$                  | $3,48 \times 10^1 \text{ J}$                       | $8,4 \times 10^{21}$ | —                                                 |
| 4–7      | ...                                               | ...                                                | $10^{18-10^6}$       | —                                                 |
| <b>8</b> | <b><math>2,16 \times 10^{-8} \text{ m}</math></b> | <b><math>1,47 \times 10^{-18} \text{ J}</math></b> | <b>354</b>           | <b>Exciton, Bohr radius, transistor threshold</b> |
| <b>9</b> | $1,62 \times 10^{-4} \text{ m}$                   | $1,95 \times 10^{-22} \text{ J}$                   | <b>0,047</b>         | 0,16 mm                                           |
| 10       | $1,21 \times 10^0 \text{ m}$                      | $2,61 \times 10^{-26} \text{ J}$                   | $6 \times 10^{-6}$   | $\sim 1 \text{ m}$                                |

**The gap across the p-bit regime**

The p-bit operating regime  $E_N \approx k_B T$  at  $T = 300 \text{ K}$  lies at  $N_{\text{opt}} \approx 8,66$  — *between* the integer levels  $N = 8$  and  $N = 9$ . The step factor is  $1/\xi_{\text{par}} = 7500$ : between  $E_8$  ( $354 k_B T$ ) and  $E_9$  ( $0,047 k_B T$ ) there is no integer level value that hits  $k_B T(300 \text{ K})$ .

**Consequence:** The  $\xi$ -energy hierarchy  $E_N$  does *not* directly fix the barrier height of the p-bit. It determines the **geometric scale** of the nanostructure at which p-bit-capable resonators naturally occur.

For cryogenic p-bits, however:  $E_9 \approx k_B T$  at  $T^* \approx 14 \text{ K}$ . At this temperature a macroscopic resonator of dimension  $L_9 \approx 0,16 \text{ mm}$  lies exactly on a hierarchy level.

**.4 The Physical Barrier: Stoner-Wohlfarth Energy****Derivation**

A magnetic nanoresonator with uniaxial anisotropy possesses an energy barrier between spin-up and spin-down, given in the Stoner-Wohlfarth model [4] by:

$$U_{\text{sw}} = K_u \cdot V \quad (5)$$

Here  $K_u$  is the uniaxial anisotropy constant (in  $\text{J/m}^3$ , material-specific) and  $V$  is the volume of the magnetic free layer.

In the FFGFT framework,  $U_{\text{sw}}$  is the macroscopic manifestation of the torsion barrier between the two stable configurations that Doc. 178 identifies as geometrically distinguished states.

For a spherical nanoresonator with diameter  $D$ :

$$U_{\text{sw}}(D) = K_u \cdot \frac{\pi}{6} D^3. \quad (6)$$

The condition for p-bit operation  $U_{\text{sw}} \approx k_B T$  fixes a specific diameter  $D^*$  for a given  $K_u$ :

$$D^* = \left( \frac{6 k_B T}{\pi K_u} \right)^{1/3}. \quad (7)$$

### Numerical values and comparison with $L_8$

For superparamagnetic nanoparticles with  $K_u \approx 10^4 \text{ J/m}^3$ :

| $D \text{ (nm)}$ | $U_{\text{sw}} \text{ (meV)}$ | $U_{\text{sw}}/k_B T_{300}$ | Regime |
|------------------|-------------------------------|-----------------------------|--------|
| 5                | 4,1                           | 0,16                        |        |
| 8                | 16,7                          | 0,65                        | p-bit  |
| 10               | 32,7                          | 1,26                        | p-bit  |
| 12               | 56,5                          | 2,18                        | p-bit  |
| 15               | 110                           | 4,3                         | p-bit  |
| 20               | 261                           | 10,1                        | p-bit  |
| 25               | 511                           | 19,8                        |        |

The p-bit operating range ( $U/k_B T \approx 1\text{--}10$ ) lies at  $D \approx 8\text{--}20 \text{ nm}$  — just below the  $\xi$ -hierarchy level  $N = 8$  with  $L_8 \approx 22 \text{ nm}$ . With Eq. (6) and  $K_u = 10^4 \text{ J/m}^3$ ,  $T = 300 \text{ K}$  one has  $U_{\text{sw}}(22 \text{ nm}) = 13.5 k_B T$  and  $U_{\text{sw}}(21.6 \text{ nm}) = 12.7 k_B T$ , i.e. above the range 1–10.

#### Geometric prediction: $N = 8$ as natural p-bit anchor

The FFGFT predicts: magnetic nanoresonators on the  $\xi$ -level  $N \approx 8$  are natural p-bit candidates at room temperature. Not because  $E_8 \approx k_B T$  (that is not the case:  $E_8 = 354 k_B T$ ), but because  $L_8 \approx 22 \text{ nm}$  is of the order of the scale at which the Stoner-Wohlfarth volume with typical material values falls into the p-bit regime; at  $K_u = 10^4 \text{ J/m}^3$ ,  $L_8$  lies just above this range, and the exact position depends on the material through  $K_u$ .

This proximity is not accidental in the FFGFT:  $N = 8$  is the universal scale of magnetic single resonators (Doc. 173: minimal resonator, Doc. 178: permanent magnet as collective resonator at  $N \approx 8$ ).

## .5 The Kramers Switching Rate in the FFGFT Framework

### Standard formula

The thermally activated switching rate between torsion configurations follows the Kramers formula [5, 6]:

$$\Gamma = \frac{\omega_0 \omega_b}{2\pi\gamma} \exp\left(-\frac{U_{\text{sw}}}{k_B T}\right), \quad (8)$$

where  $\omega_0$  is the angular frequency at the potential minimum,  $\omega_b$  at the barrier peak and  $\gamma$  is the (Gilbert) damping rate.

### FFGFT interpretation: determinism instead of randomness

In the standard interpretation,  $\Gamma$  is a stochastic rate: each transition is a random, uncorrelated event.

### FFGFT reinterpretation of the Kramers rate

In the FFGFT, equation (8) is **formally identical** — but its physical meaning changes:  $\Gamma$  is the **time-averaged frequency of deterministic torsion transitions**. Each individual transition follows a precise path in the sub-Planck-scale geometry of the torsion field, which is determined on the timescale  $t_0 = t_p/7500$ .

The macroscopic measurement — temporal resolution  $\Delta t \gg t_0$  — averages over  $\Delta t/t_0 \gg 1$  microstates and creates the appearance of randomness. The randomness is **epistemic**, not ontological.

This is the direct transfer of the FFGFT position established in Doc. 178 (epistemic ignorance rather than genuine superposition) to the thermally fluctuating p-bit system.

### Why no modified prefactor

A numerical FFGFT correction to the Kramers rate would require the structural parameter  $k_B T t_0 / \hbar$ :

$$\frac{k_B T t_0}{\hbar} = \frac{k_B \cdot 300 \text{ K}}{\hbar/t_0} \approx 2,82 \times 10^{-34}. \quad (9)$$

This value is  $\ll 1$  at all technically accessible temperatures. A correction to the Kramers prefactor by the FFGFT is not experimentally detectable at room temperature.

### Where the FFGFT contribution really lies

1. **Geometric scale:**  $D \sim L_8 \approx 22 \text{ nm}$  as natural p-bit anchor (order of magnitude; the exact position depends on the material through  $K_u$ ) — verifiable materials-science prediction.
2. **Interpretation:** The switching statistics are deterministically structured on the  $t_0$  scale; the macroscopically observed stochasticity is a projection.
3. **No new terms:** The Kramers rate (8) remains unchanged. The FFGFT contribution is interpretive.

## .6 The Three-Regime Picture: Bit, p-Bit, Qubit

| Type         | Barrier               | Temp. | Timescale   |
|--------------|-----------------------|-------|-------------|
| Class. bit   | $U \gg k_B T$         | 300 K | Years       |
| <b>p-bit</b> | $U \approx k_B T$     | 300 K | ns– $\mu$ s |
| Qubit        | $U \gg k_B T$ (prep.) | mK    | $\mu$ s–ms  |

| Type         | $\xi$ -level                       | FFGFT: random?           |
|--------------|------------------------------------|--------------------------|
| Class. bit   | $N \approx 8$ , deep               | No; stable torsion well  |
| <b>p-bit</b> | $D \sim L_8 \approx 22 \text{ nm}$ | No; deterministic        |
| Qubit        | $N \approx 8$ , metastable         | No; epistemic (Doc. 178) |

All three types reside on the same geometric scale  $N \approx 8$ . The difference is the operating point on the potential curve: deep in the minimum (bit), at the tipping point (p-bit), metastably prepared (qubit).

## .7 Landauer's Principle

Landauer's principle [7] applies to the p-bit unchanged:

$$E_{\text{diss}}^{\min} = k_B T \ln 2. \quad (10)$$

The p-bit switches between two torsion configurations; the state space is binary. The  $\xi$ -geometry determines the *dynamics* (scale, switching rate), not the *state space size*.

### No modified Landauer bound in the FFGFT

A  $\xi$ -dependent Landauer bound would require the state space of the p-bit to have more than two elements, or that information erasure uses geometrically non-equivalent paths. Neither is the case in a standard p-bit. The standard bound  $k_B T \ln 2$  applies.

## .8 Experimental Realisations

- **Magnetic Tunnel Junctions (MTJ) [2]:** CoFeB free layers,  $K_u \sim 10^4\text{--}10^5 \text{ J/m}^3$ , lateral dimension  $\sim 20\text{--}100 \text{ nm}$  — directly at  $L_8$ . Switching times in the nanosecond range.
- **Superparamagnetic nanoparticles:**  $\text{Fe}_3\text{O}_4$ ,  $\gamma\text{-Fe}_2\text{O}_3$  with  $D \approx 10\text{--}20 \text{ nm}$  show spontaneous thermal switching at 300 K in the  $L_8$  regime.
- **CMOS-based p-bits [3]:** Ring oscillators near metastability. No direct torsion field connection — realise the logical operating point by other means.

### Verifiable FFGFT prediction

The FFGFT names  $L_8 \approx 22 \text{ nm}$  as a **geometrically distinguished dimension** for p-bits. Eq. (7) with  $K_u = 10^4 \text{ J/m}^3$  and  $T = 300 \text{ K}$ , however, gives  $D^* = 9.2 \text{ nm}$ , not  $L_8$ ;  $D^*$  depends on the material through  $K_u^{-1/3}$ . That  $D^*$  coincides with  $L_8$  therefore does not follow from the equation but requires a material with a suitable  $K_u$ . A systematic measurement of switching rate, noise colour and reproducibility across different nanoparticle sizes would show whether an optimum occurs at  $D \approx L_8$ .

## .9 Classification within the FFGFT Document Series

| Doc.       | Topic                                            | Relation to Doc. 184                                                           |
|------------|--------------------------------------------------|--------------------------------------------------------------------------------|
| 173        | $\xi$ -hierarchy, quantum dot, minimal resonator | Foundation: $L_N$ , $E_N$ derivation, $N = 8$ anchor                           |
| 174        | Reading and writing spin                         | Techniques; decoherence times                                                  |
| 178        | Spin stability, permanent magnet                 | Epistemic randomness; torsion well vs. tipping point                           |
| <b>184</b> | <b>p-bit</b>                                     | <b>Tipping point: same scale <math>N = 8</math>, different operating point</b> |

## .10 Summary

### Conclusion: p-bit in the FFGFT framework

**1. The  $\xi$ -energy hierarchy is cleanly derivable.**

$E_N = \hbar c / L_N = E_P \cdot \xi_{\text{par}}^{N-1}$ , with  $E_1 = E_P$ . The levels jump by a factor of 7500 — the p-bit regime at 300 K lies between  $N = 8$  and  $N = 9$ .

**2. The barrier is Stoner-Wohlfarth:  $U = K_u \cdot V$ .**

Adjustable by materials engineering, not quantised by  $E_N$ . p-bit regime at  $D \approx 8\text{--}20\text{ nm}$  ( $K_u \sim 10^4\text{ J/m}^3$ ); the exact position depends on the material through  $K_u$ .

**3. The  $\xi$ -reference is the geometric scale  $L_8 \approx 22\text{ nm}$ .**

This is the universal resonator scale  $N = 8$  from Doc. 173. Natural p-bit candidates lie near this level, just below it at  $K_u = 10^4\text{ J/m}^3$ .

**4. The Kramers rate remains formally unchanged.**

FFGFT contribution: determinism of transitions on the  $t_0$  scale — the stochasticity is epistemic, not ontological.

**5. The Landauer bound applies unchanged.**

$E_{\text{diss}}^{\text{min}} = k_B T \ln 2$  — binary state space, no  $\xi$  modification.

**The p-bit is the geometric tipping point between permanent magnet (Doc. 178) and qubit (Doc. 174) — on the same  $\xi$ -level  $N = 8$ .**

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